

FISI4997: Rotation Curve of the Milky Way

IVARIS C. MARTÍNEZ-SERRANO¹

¹*University of Puerto Rico
Mayagüez, PR, USA*

1. THE INNER GALAXY

The goal of this part of the project is to obtain neutral hydrogen (HI) 21-cm spectra from different positions in the inner region of the Milky Way and use them to derive radial velocities. These measurements will later be used to construct a galactic rotation curve. The HI 21-cm line arises from the hyperfine transition of neutral hydrogen and is a powerful tracer of interstellar gas throughout the galaxy (Ostlie & Carroll (1996), Section 12.1). Because this radiation is not strongly absorbed by dust, it allows us to probe regions across the galactic plane, including distant structures in the inner galaxy.

1.1. Data Acquisition

To focus on the inner galaxy, a set of eight positions within the galactic longitudes $l = 270^\circ$ to $l = 360^\circ$ and galactic latitudes $b = \pm 1.5^\circ$ was chosen, using Aladdin Baumann et al. (2022). This range of longitudes seemed to go through the inner galaxy, where the tangent point method (Ostlie & Carroll (1996), Section 24.3) can be applied to relate observed radial velocities to rotational velocities. Using the HI4PI Survey Data Interface HI4PI Collaboration et al. (2016), I decided to go from around $l = 280^\circ$ to $l = 350^\circ$ in steps of around 10° , varying the b values slightly, trying to get spectra relatively easy to fit. At the end, I chose the values shown in Table 1. The smallest possible beam size (0.2°) was chosen, to minimize blending of multiple gas components along the line of sight. The velocity range was also left at its default setting to capture the full distribution of radial velocities.

A screenshot of each spectrum was taken for reference (super helpful) and then each spectrum was downloaded as a *.txt* file. The spectral profiles are shown in Figure 1.

When observing HI emission across the galaxy, the expected range of radial velocities depends on the direction of observation. Radial velocities in the inner galaxy can

Table 1. Observed HI Spectra Positions

Spectra	Longitude (l°)	Latitude (b°)
1	277.0	-1.0
2	291.0	-1.2
3	301.0	-0.2
4	311.0	0.0
5	318.0	-1.5
6	328.0	1.0
7	339.0	0.0
8	350.0	0.4

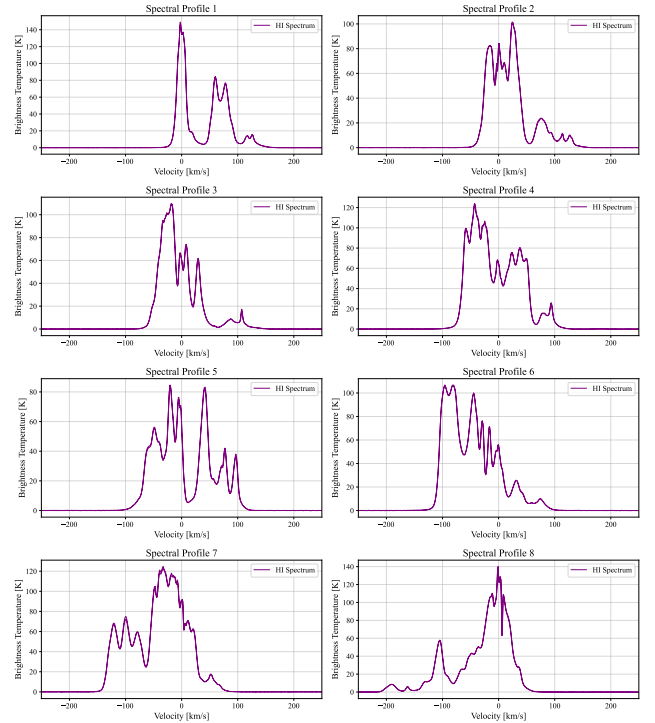


Figure 1. HI 21-cm spectra for the selected galactic coordinates listed in Table 1. Each spectrum shows brightness temperature as a function of radial velocity. Each peak corresponds to a neutral hydrogen cloud along the line of sight. Higher absolute velocities are supposed to trace the gas deeper within the inner galaxy.

ential rotation and projection effects. Because of this, I chose to fit only the narrow, well-separated peaks at the highest velocities. Emission near $v_r = 0$ km/s was ignored, because they are typically associated with local gas and is not useful for studying galactic rotation.

2. FITTING THE SPECTRA

The HI spectra shown in Figure 1 was analyzed by fitting Gaussian models to the observed emission lines. This was done using a Python code that has an interactive selection of spectral features. This code returns the best-fit parameters for each Gaussian component along the region selected. Each Gaussian fit provided three key parameters:

- The amplitude (A), which corresponds to the peak brightness temperature.
- The central velocity (V_{LSR}), which represents the radial velocity of the gas component relative to the Local Standard of Rest (position of our Sun).
- The full width at half maximum (FWHM), which shows the velocity dispersion of the gas.

For the purpose of constructing the galactic rotation curve, the most important parameter is the central velocity, because with it we can estimate the radial velocity along each line of sight. The Gaussian fits of each spectra are shown in Figure 2.

Many of the HI spectra showed multiple peaks corresponding to gas at different distances along the line of sight. Here, we made the assumption that the highest absolute velocity components correspond to gas located at the tangent point, where the line of sight is tangent to a circular orbit around the galactic center, so the Gaussian fits were only applied to the highest absolute velocity components in each spectrum. The Gaussian fit parameters of the highest absolute velocity components of each spectral profile are shown in Table 2.

Table 2. Gaussian Fit Parameters of the Highest Absolute Velocity Components of our HI Spectra.

Spectra	Amplitude (err)	Gaussian Center (err)	FWHM (err)
1	10.75(0.44)	122.95(0.33)	18.21(0.95)
2	9.22(0.18)	127.38(0.14)	13.13(0.34)
3	11.01(0.38)	107.36(0.06)	3.58(0.15)
4	14.38(0.42)	94.04(0.06)	4.66(0.17)
5	35.06(1.08)	96.68(0.15)	9.36(0.37)
6	57.09(1.60)	-99.74(0.07)	12.51(0.29)
7	67.12(0.21)	-121.18(0.05)	20.15(0.12)
8	8.44(0.058)	-191.31(0.07)	19.95(0.16)

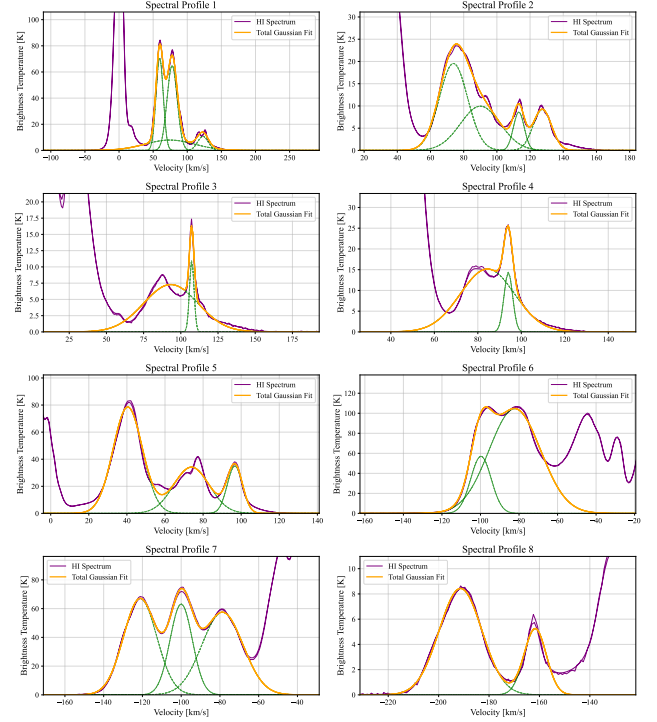


Figure 2. Gaussian fits to the HI 21-cm spectra. The fitted models (overlaid on the spectral profiles) were used to get the central velocity (V_{LSR}), amplitude, and linewidth (FWHM) for each selected spectral component.

2.1. Deriving the Rotation Velocity

If we assume circular motion of gas around the galactic center, the observed radial velocity can be related to the angular velocities of the gas and the Sun (Ostlie & Carroll (1996), Section 24.3). We can use Equation (24.37) to calculate relative radial velocity:

$$v_r = (\Omega - \Omega_0)R_0 \sin \ell, \quad (1)$$

Where R_0 is distance our Sun has from our galaxy's center. From this expression, we can substitute the definitions of angular velocity:

$$\Omega = \frac{\Theta}{R} \text{ and } \Omega_0 = \frac{\Theta_0}{R_0}, \quad (2)$$

where Θ_0 is the Sun's orbital velocity, and now convert the equation into:

$$v_r = \left(\frac{\Theta}{R} - \frac{\Theta_0}{R_0} \right) R_0 \sin \ell, \quad (3)$$

$$v_r = \frac{\Theta R_0 \sin \ell}{R} - \Theta_0 \sin \ell. \quad (4)$$

But from the Figure 24.22, we can see that the geometry between our Sun, the gas component and the center of our galaxy gives:

$$R = R_0 \sin \ell. \quad (5)$$

We call this the tangent point method. So the equation now becomes:

$$v_r = \frac{\Theta(R_0 \sin \ell)}{R_0 \sin \ell} - \Theta_0 \sin \ell \quad (6)$$

$$v_r = \Theta - \Theta_0 \sin \ell. \quad (7)$$

We can now solve for the rotational velocity, and get:

$$\Theta(R) = v_r + \Theta_0 \sin \ell, \quad (8)$$

and using the orbital velocity of the Sun Θ_0 as $\Theta_0 = 220$ km/s from Equation (24.36) we get:

$$\Theta(R) = v_r + (220) \sin \ell, \quad (9)$$

Using equation (5), where $R_0 = 8$ kpc from Equation (24.8), and equation (9) we got the radius and rotational velocities of our fitted components. We show them in Table 3. We plotted the values shown in Table 3 to

Table 3. Radius and Rotational Velocities of our Fitted Gas Components

Spectra	R [kpc]	$\Theta(R)$ [km/s]
1	7.94	341.31 ± 0.33
2	7.47	332.77 ± 0.14
3	6.86	295.94 ± 0.06
4	6.04	260.08 ± 0.06
5	5.35	243.89 ± 0.15
6	4.24	216.32 ± 0.07
7	2.87	200.02 ± 0.05
8	1.39	229.51 ± 0.07

give a simple galactic rotational curve, which we show in Figure 3.

From the image, we can see that the rotational velocity trend is to increase as the galactocentric radius increases, but it seems to decrease slightly from 1 kpc to 3 kpc. This trend behavior is consistent with what we would expect for gas in the inner Milky Way. However, sources of error are always present; these can come from:

- Errors in the Gaussian fit parameters.
- The use of Gaussian centers instead of true terminal velocities.
- Deviations from perfectly circular motion.
- Line-of-sight blending of multiple gas components.

Then the smallest velocity point may be affected by these errors the most as indicated by its deviation from the general trend.

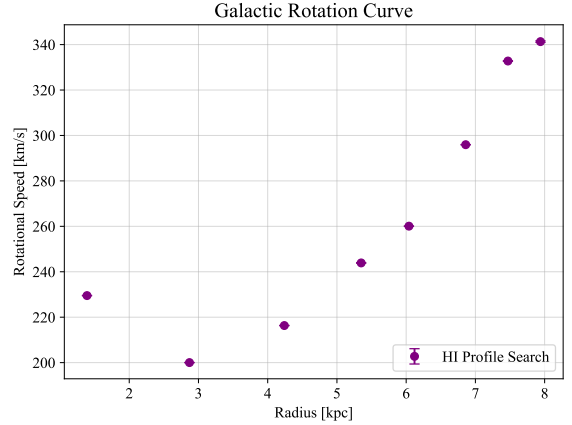


Figure 3. Galactic rotation curve derived from HI 21-cm observations using the tangent point method. The rotational velocity $\Theta(R)$ is plotted as a function of galactocentric radius R .

3. THE OUTER GALAXY

The method we used in Part 2 is only valid over a limited range of galactic longitudes ($0^\circ < \ell < 90^\circ$ and $270^\circ < \ell < 360^\circ$), which correspond to lines of sight that pass inside the Sun's orbit around the Galactic center. However, in the outer galaxy ($90^\circ < \ell < 270^\circ$), the line of sight does not intersect a tangent point. Instead, it passes only through regions outside the Sun's orbit. Because of this, we need to make measurements of distance to construct the rotation curve in this outer region. To do this, we used the WISE HII Survey (Anderson et al. 2014) database. This database shows the information of regions of ionized hydrogen clouds associated with star-forming regions. With these, we can create an outer galaxy rotation curve because their distances and radial velocities have been previously measured. I decided to download the data from Table 2 and Table 6, which had information about these ionized gas structure's velocity and distances, respectively. For each source, we used their galactic longitudes (ℓ) and radial velocities (v_r), to get their galactocentric radius (R) and rotational velocity (Θ). For the inner galaxy once again, we can do as we did in Part 2, taking the following equation:

$$R = R_0 \sin \ell, \quad (10)$$

to get their galactocentric radius, and the following equation:

$$\Theta(R) = v_r + (220) \sin \ell, \quad (11)$$

to get their rotational velocities. For the outer galaxy, we need to use a different methodology. For this region, the galactocentric radius (R) can be taken directly from the catalog with the column $RGal$, but the rotational

velocity has to be calculated using a different equation:

$$\Theta(R) = \left(\left| \frac{v_r}{\sin \ell} \right| + \Theta_0 \right) \frac{R}{R_0}, \quad (12)$$

where the absolute value needs to be there because radial velocities can be positive or negative. We have to keep in mind that these calculations once again assume perfect circular motions of the gas in the HII regions. We show the resulting galaxy rotation curve in Figure 4.

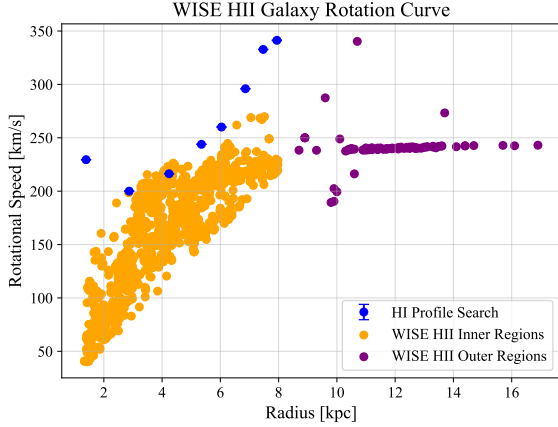


Figure 4. Galactic rotation curve from WISE HII data along with the HI data. The new WISE data extends the rotation curve beyond our solar radius.

Including this data allows us to create a rotation curve that is more extended to larger galactocentric radii beyond our Sun's. The inner region data points show a trend relatively consistent with the data points from the HI profile searches, although these former ones do seem to show higher velocities than the WISE data. The rotation curve remains relatively flat at large radii, which is what we would expect if we are taking into account the presence of dark matter in the Milky Way.

4. COMPARISON WITH OTHER DATA

In a system where most of the mass is concentrated near the center (such as the Solar System), we expect the rotational velocity to follow a Keplerian profile, which has the form:

$$\Theta(R) \propto \frac{1}{\sqrt{R}}, \quad (13)$$

If our galaxy were to follow this form, the rotational velocity of the gas components in it should decrease with increasing radius. To compare, we have added this keplerian model to our data plot in Figure 5. From this image, we can observe that the rotation curve of our galaxy does not follow this keplerian form. Instead the velocities remains flat at large radii. This indicates that

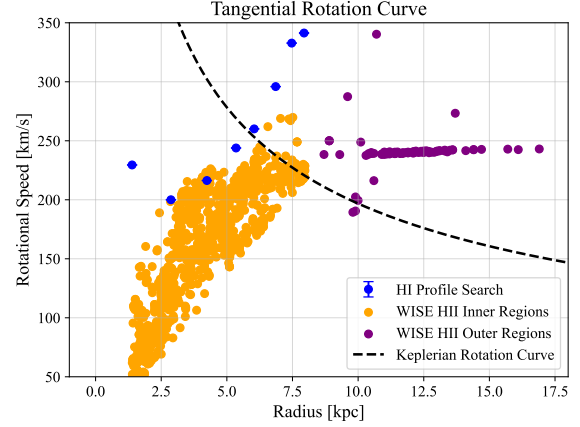


Figure 5. Our galactic rotation curve plotted with a keplerian rotation curve for comparison.

the mass of the galaxy is not concentrated at its center, and that it continues to increase along its radius. Scientists believe the explanation for this is dark matter, which would surround the galaxy and act as additional gravitational mass that is not directly observable. Another fun thing we can do with this plot is to compare it to results from peer-reviewed papers. In Figure 6 we have done just that.

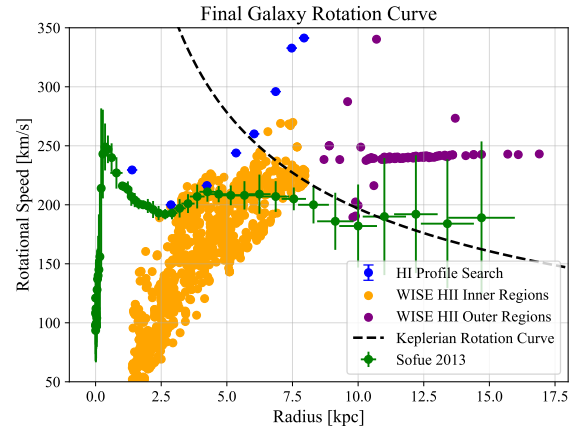


Figure 6. Final galactic rotation curve compared with data from Sofue (2013).

The general trend of the rotation curve is at large radii seems consistent with our data, we can see that their rotation curve becomes flat at these distances. However, at small radii, there seems to be some inconsistencies. Their inner galaxy shows a trend that sharply peaks and then slightly falls to a flat structure, while our inner galaxy data points do not show this at all. Instead, our data points consistently go up in velocity the farther away they are from the center of our galaxy. Another

thing we can observe from this comparison is that our measured velocities near the flat line are higher than what is shown in Sofue (2013)’s paper and our data seems much more scattered than theirs. These differences maybe arise from our use of Gaussian centers instead of real terminal velocities. Or maybe our tiny sampling along all galactic positions. And maybe human error when doing the measurements, calculations and fittings. However, even though these differences are real, it seems like they overall agree with each other at large radii, which is arguably the most important part. This then supports what we see with our data measurements and also confirms that the really interesting flat feature we explained with dark matter is real and that other people have measured it also.

4.1. *Uncertainties and Sources of Error*

There exists many sources of uncertainty in our rotation curve. Perhaps the biggest one being the assumption of circular motion. Our analysis relied heavily on the fact that all gas follows perfectly circular orbits. In reality this is much different and non-circular motions, like how spiral arms move, probably introduce many variations. Another thing we need to consider are possible distance uncertainties in the outer region, because a tiny error in this distance can change the curve a lot. We are heavily trusting that the distances given from the WISE survey are correct. We can also consider the error that gaussian fits can introduce. As we

established before, the central velocities that gaussian fits give us may vary wildly from the velocities at the edge of the gaussian. The true terminal velocity lies at the edge of the emission profile and is often slightly larger in magnitude, but it may get lost with using the fits parameters. Other assumptions, like the Sun’s position or the galactic longitude measurements of WISE may introduce errors. We plotted the error bars in our figures using the uncertainties in the Gaussian center velocities. However, these probably undermine the true total uncertainty that comes from the things said above.

4.2. *Other Galaxies*

We can apply this type of analysis to other galaxies, but we need to be careful on how we do it. For example, for spiral galaxies we can use Doppler shifts of emission lines from across the galaxy disk. This is something we cannot do easily in the Milky Way. Far away galaxies allow us to directly map its rotational velocity as a function of radius. For elliptical galaxies, this get a bit more complicated, because these galaxies lack significant hydrogen gas that would help us study it. Instead we can use things like stellar velocity dispersions, absorption line spectroscopy, and X-ray observations of hot gas (Rubin et al. (1978)).

5. CODE

All of the codes and data used to produce these images can be found in this [GitHub](#).

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